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Relationship and personality factors predict longitudinal changes in dream content

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Researchers have established that dreams are intensely social and populated by diverse characters, including important figures from the dreamer's daily life. This study examines the types of characters that appeared in participant dreams over two weeks. We found that the majority of dreams include strangers in addition to known individuals, and that personality measures impact the likelihood of dreaming about different types of people. Appearance of known individuals from daily life in dreams was assessed by comparing dream reports to the core support networks of participants and daily diaries. We found that relationship-specific variables and daily interaction were important predictors of the likelihood of support network dream appearances. While daily interaction generally increases the likelihood of dream appearances, this effect is reversed for important family members like parents or siblings, indicating that dreams may play a compensatory role in maintaining relationships with close others when they are not present.

The problem of the potential adaptive function of dreams is now firmly on the scientific agenda as the experimental tools, data sources, and conceptual paradigms available to address the issue have all increased in precision in the last few years[1–4]. In spite of this growth in research, however, it remains an open question what the specific adaptive function of dreams are, or if they even have a function at all[5]. Experimentally supported hypotheses on the adaptive functions of dreams include: *Simulation of counterfactual virtual worlds*[6–8] ; *cognitive model updating*[9] within a Bayesian brain or predictive processing theoretical framework; *problem solving and creativity*[10,11]; *threat simulation*[12], which via practice effects would enhance responses to daytime threats; *emotional regulation* through linking emotional events with less-distressing contexts[1,13]; and *social simulation*, which simulates social interactions with individuals important to the fitness of the dreamer. It is this last class of theories, the social simulation functions of dreams, that we tested in the series of studies presented in this paper.

Dreams are intensely social and are populated by a wide variety of social characters, ranging from those the dreamer knows well in real life to completely fabricated characters with no waking analogue[14]. Categorizing these dream characters has long been of interest to researchers[15], and content analysis studies have found that people frequently dream about romantic partners, family members, and people they interact with[16–18].

The ubiquity of social interactions in dreaming forms the basis of the Social Simulation Theory (SST), which contends that the function of dreaming is to process and update cognitive and emotional schemas of social interaction[19]. Revonsuo and Tuominen[19] note that 95% or more of dreams are populated, with dreamers interacting with two to four other characters, some of whom can be recognized as familiar characters in the dreamer's immediate social network. Friendly interactions (typically verbal conversations) are found in about 40% of dreams, while aggressive social interactions occur in about 45% of dreams. In addition, mind reading or inferring the mental states of others, particularly those characters the dreamer interacts with, occurs in more than 80% of dreams. According to the Continuity Hypothesis (which is a descriptive account of dream content that does not posit a functional account of dreams), however, dreaming contains a large variety of social interactions as a byproduct of the sociality of waking life[17,20]. Different articulations of the Continuity Hypothesis vary in whether they focus only on events from daily life[21] or also reflect cognitive processes such as thoughts, preoccupations, and emotions[20]. A key division between Social Simulation Theory and the Continuity Hypothesis is whether or not dreams are social "rehearsals" that allow for greater function in daily life or whether they are merely reenactments or dramatizations of waking concerns[22]. Running parallel to this debate is work on the interconnection between dreaming and social attachment, which has found that attachment style (e.g. avoidant, pre-occupied, or anxious) influences levels of dream recall[23], dream content[24,25], and that dream content can influence waking attitudes towards romantic partners[26].

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To assess whether social simulations in dreams are more than mere reflections of everyday social interactions and are instead functional rehearsals of difficult social interactions, Tuominen, Olkoniemi, Revonsuo, and Valli[27] analyzed the dreams of participants ($N = 18$) before, during, and following a period of social isolation. If social simulations are merely reflective of everyday interactions, then those simulations should disappear to some extent during the isolation period. They found that dreams in seclusion still showed high levels of sociality, indicating that dreams have a social bias even when not pressured by typical daily social interactions. More intriguingly, they also found that seclusion dreams included higher levels of familiar characters (family, friends, and romantic partners), which Tuominen and colleagues[27] contend indicates that dreams provide a means for maintaining and strengthening attachment bonds when there is less opportunity for attending to them in waking life. The role of dreams in facilitating these bonds may vary according to time and relationship function, however. A study on dreaming of partners and ex-partners by Schredl, Cadiñanos Echevarria, Saint Macary, and Weiss[28] found that ex-partner dreams declined depending on both length of time since that relationship and the length of any new partner relationships.

In the following set of studies, we reasoned that if dreams simulate social interactions and facilitate attachments or social bonding in the waking world, then the personality and attachment structure of the individual should influence the content of dream simulations. Personality traits are known to significantly influence quality and variety of social relationships[29] and indeed, dream content itself[30]. Remarkably, however, there have been few studies examining the impact of personality traits on social simulations in dreams (although see König and colleagues' 2016 work[31]). It is important to do so as this effect is a strong prediction of social simulation theory. If we find that personality and attachment traits such as extraversion or anxious attachment strongly predict content of social simulations in dreams including character numbers, types and interactions, then confidence in the attachment component of social simulation theory would increase.

In addition to the above-mentioned partial failure to address personality and attachment-related variables in relation to the SST, the fine-grained predictions of the SST, such as the appearance of differing kinds of characters, have not yet been adequately addressed. Most importantly, in order to assess differences in predictions of the continuity hypothesis, the attachment hypothesis and the social simulation hypothesis, the literature has not yet addressed the incorporation of identifiable individuals either named by participants as part of their core support network or in their descriptions of their daily activity. As we noted above, if continuity is correct then characters who are part of the dreamer's core social network should appear regularly in their dreams. However, if SST governs character appearance then only those characters who are presenting a social challenge to the dreamer should appear regularly. Whether or not a core network character presents a social challenge to the dreamer very likely depends on day to day interactions with these individuals. Unfortunately, to our knowledge, few or no studies on SST have examined day to day interactions between core network characters and the dreamer.

Given the above existing limitations of the existing literature on SST we designed the following studies to summarize the appearance of dream characters over two weeks in an adult community-dwelling sample. First, we analyze the appearance of different kinds of characters in dream content in relation to personality and attachment style. We then turn to dream content analysis to assess the incorporation of identifiable individuals either named by participants as part of their core support network or in their descriptions of their daily activity. Finally, we assess the impact of relationship-specific factors and daily interactions on the likelihood of dream appearances of individuals from participant's core support network.

This study provides a novel contribution to the study of the social aspects of dreaming by gathering longitudinal data from a relatively large sample of adults in the home, as opposed to retrospective or cross-sectional measures. In addition, to our knowledge we are the first study to measure the influence of multiple relationship-specific variables on dream appearance over time, which provides greater insight into the role of relationship quality and frequency of interaction. Our dataset thus provides a unique vantage point for examining key open questions in the social dimensions of dreaming.

Results

Dream population predicted by baseline personality measures

We tested the likelihood of certain types of characters being present in participant ratings as a function of baseline personality and attachment. First, we calculated the number of co-occurrences of character types in dreams (Fig. 1). Strangers occurred the most overall, followed by Friends and Relatives. The highest level of co-occurrences was Strangers and Friends (Jaccard Index = 0.21) followed by Friends and Relatives (Jaccard Index = 0.22).

We next fit a series of logistic mixed effects models to test the effects of personality (measured via the Big Five Inventory; BFI) and baseline attachment (measured by the Adult Attachment Scale; AAS) on the likelihood of certain kinds of dream characters to appear in dream content (Table 1; for full model outputs see Table S1 in Supplementary Materials, see Sect. 4 for references on BFI and AAS). Significance was assessed after conducting the Benjamini-Hochberg procedure to adjust for multiple comparisons.

This series of models found that higher levels of neuroticism predict the presence of relatives in dreams, while friends are positively predicted by extraversion. The likelihood of strangers appearing in dreams is negatively influenced by extraversion but positively influenced by openness.

Daily life and ego network characters in dreams

We next examined the appearances of individuals in dreams that were known to the participants, either from their core support network or individuals not in this network that they reported interacting with in daily life. We first report the overall numbers of dream nights without any identifiable individuals, dream nights with at least one of either type of individual, and dream nights with at least one of both types of individuals (Table 2).

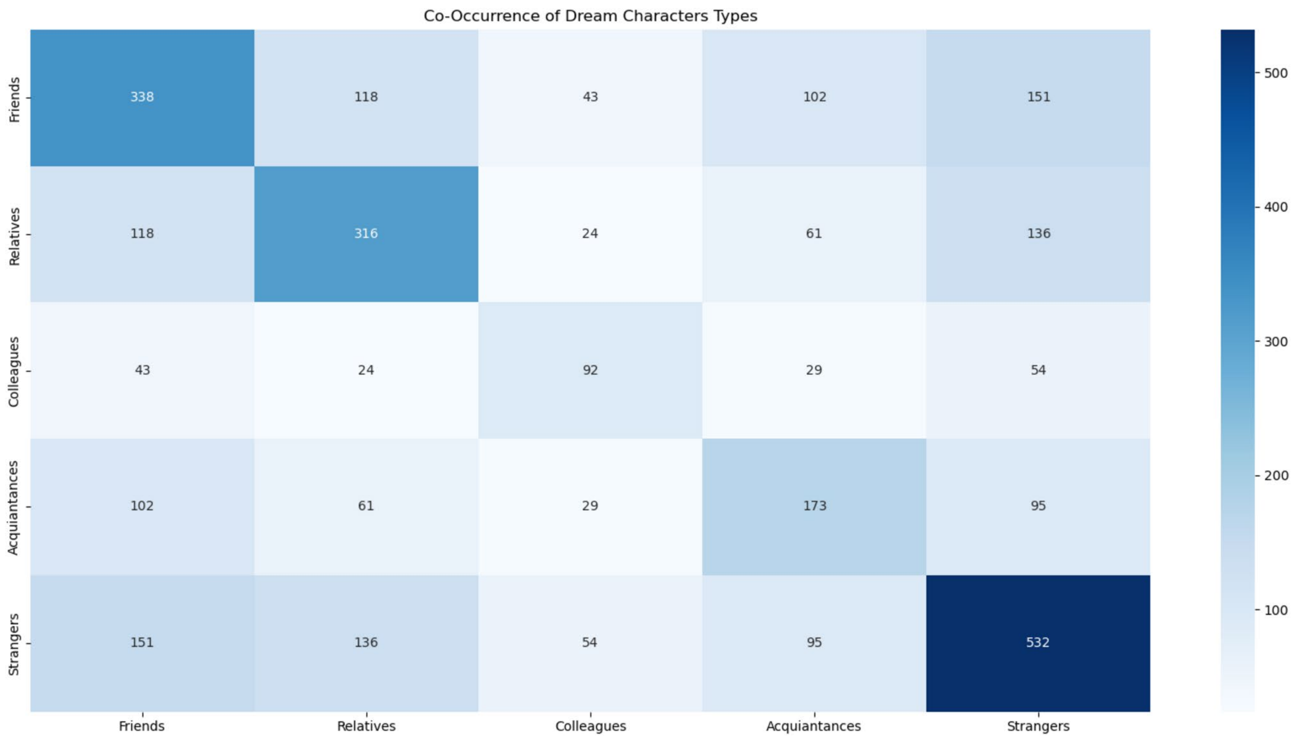

Fig. 1. Co-Occurrence of dream character types. This heatmap summarizes the appearance of different characters during dream nights in binary ratings by participants (1 = Character type present). Dream nights in which only that character type appeared are on the diagonal, with co-occurrences appearing in their respective cells.

Predictor	Relatives	Friends	Acquaintances	Colleagues	Strangers
BFI-E	0.048	0.404**	0.292	−0.028	−0.494***
BFI-A	0.21	−0.122	−0.149	−0.254	−0.111
BFI-C	0.284*	−0.183	−0.108	0.169	0.049
BFI-N	0.55***	0.011	−0.168	0.028	−0.066
BFI-O	−0.237	−0.129	0.042	0.379	0.45***
AAS-ANX	−0.268*	0.263	0.097	−0.235	−0.03
AAS-AVO	−0.08	−0.166	0.116	−0.047	−0.045

Table 1. Logistic mixed effects model coefficients predicting dream characters from personality and attachment. This table summarizes the outputs of a series of models predicting participant ratings of the presence of dream character types as a function of personality and attachment. BFI = Big five inventory: e = extraversion, a = agreeableness, c = conscientiousness, n = neuroticism, o = openness; aas = adult attachment. Scale: anx = anxiety, avo = avoidance. *$p < .05$, ** $p < .01$, *** $p < .001$.

In the model predicting daily character appearance, only avoidant attachment (AAS-ANX) was negative and significant (b = -0.57). In the model predicting ego network appearance, only conscientiousness (BFI-C) was positive and significant (b = 0.31).

Relationship variables influencing daily interaction and dream appearance

We first report the overall numbers of daily interactions and dream appearances depending on relationship type (Tables 5 and 6; Fig. 2). For dream interactions, we excluded all no recall nights.

We found that Partners and Parents had the highest level of dream appearances, and overall that family members in general had notably higher ratios of dream appearances than non-family members.

We next compared the influence of participant ratings of relationship variables on each type of appearance. We fit two mixed-effects models controlling for participant variability and for participant-relationship variability, since relationships were clustered within individuals (Table 7).

For daily appearances, relationship frequency was the strongest predictor ($\beta = 1.391$, $p < .001$). Closeness ($\beta = 0.284$, $p < .001$) and conflict ($\beta = 0.214$, $p < .001$) were also positive and significant, along with financial

	# of Dream Nights	Ratio
No individuals from support network or observed daily life	857	0.74
At least 1 observed from daily life	73	0.06
At least 1 from support network	195	0.17
Both daily life and support network	38	0.03

Table 2. Recognizable characters in Dreams from everyday Life. This table summarizes the number and proportion of the appearance of individuals from participants' everyday life that could be identified by the researchers either through participants naming them in their core support network or mentioning them in diaries of daily activity. The vast majority of dream nights did not include individuals that could be identified by the study team either based on the core network or daily interactions. Support network individuals appeared at a little over double the rate of individuals from daily life, with a minority of Dreams including representatives from both categories. We next fit two mixed effects logistic models testing the effects of personality and attachment predictors on the presence of support network individuals and people from daily life in Dreams (Tables 3 & 4).

						95% CI		95% Prob CI	
Outcome	**Predictor**	**β**	**Prob**	**SE**	**Z-Stat**	**Low**	**High**	**Low**	**High**
Daily Character Appearance									
	Level-1								
	Intercept	−3.348	0.034	0.294	−11.373	−3.925	−2.771	0.019	0.059
	Level-2								
	BFI-E	0.172	0.543	0.289	0.597	−0.393	0.738	0.403	0.677
	BFI-A	−0.262	0.435	0.275	−0.952	−0.801	0.277	0.31	0.569
	BFI-C	0.254	0.563	0.259	0.981	−0.253	0.761	0.437	0.682
	BFI-N	0.003	0.501	0.305	0.009	−0.595	0.601	0.355	0.646
	BFI-O	0.137	0.534	0.268	0.509	−0.389	0.662	0.404	0.66
	AAS-ANX	0.006	0.501	0.26	0.022	−0.503	0.515	0.377	0.626
	AAS-AVOID	−0.456*	0.388	0.232	−1.964	−0.91	−0.001	0.287	0.5

Table 3. Logistic mixed effects model predicting appearance of characters from daily Life. This table summarizes the output of a logistic mixed effects model predicting dream nights with characters from daily life from personality and attachment factors. BFI = Big five inventory: e = extraversion, a = agreeableness, c = conscientiousness, n = neuroticism, o = openness; aas = adult attachment scale: anx = anxiety, AVO = Avoidance. *$p < .05$. Prob. = the predicted probability from the log-odds coefficient.

						95% CI		95% Prob CI	
Outcome	**Predictor**	**β**	**Prob**	**SE**	**Z-Stat**	**Low**	**High**	**Low**	**High**
Ego Network Appearance									
	Level-1								
	Intercept	−1.806	0.141	0.151	−11.932	−2.102	−1.509	0.109	0.181
	Level-2								
	BFI-E	−0.334	0.417	0.173	−1.933	−0.674	0.005	0.338	0.501
	BFI-A	−0.224	0.444	0.173	−1.296	−0.562	0.115	0.363	0.529
	BFI-C	0.315*	0.578	0.156	2.025	0.01	0.62	0.503	0.65
	BFI-N	0.136	0.534	0.192	0.707	−0.241	0.513	0.44	0.626
	BFI-O	0.122	0.531	0.16	0.764	−0.191	0.436	0.452	0.607
	AAS-AVOID	−0.128	0.468	0.163	−0.785	−0.447	0.191	0.39	0.548

Table 4. Logistic mixed effects model predicting appearance of characters from core support Network. This table summarizes the output of a logistic mixed effects model predicting dream nights with characters from participants' core support network from personality and attachment factors. BFI = Big five inventory: e = extraversion, a = agreeableness, c = conscientiousness, n = neuroticism, o = openness; aas = adult attachment scale: anx = anxiety, AVO = Avoidance. *$p < .05$. Prob. = the predicted probability from the log-odds coefficient.

	No Daytime Interaction	Daytime Interaction	Ratio
Partner	200	934	0.82
Parent	1008	434	0.3
Sibling	844	234	0.22
Child	622	554	0.47
Other Relative	868	168	0.16
Friend	2746	754	0.22
Ex-Partner	80	32	0.29
Colleague	480	150	0.24
Schoolmate	82	16	0.16
Member of Religious Group	116	10	0.08
Neighbor	188	36	0.16
Other Relative	159	37	0.19

Table 5. Ratio of daily interactions with core support network members by relationship Type. This table compares the number of observed interactions for all 124 participants over a two-week period with members of participants' core support networks summarized by relationship type. Participants were asked if they interacted with each network member each day and indicated yes or no.

	Not in Dream	In Dream	Ratio
Partner	621	107	0.15
Parent	860	85	0.09
Sibling	690	27	0.04
Child	676	44	0.06
Other Relative	696	4	0.01
Friend	2361	38	0.02
Ex-Partner	92	3	0.03
Colleague	466	1	0
Schoolmate	82	0	0
Member of Religious Group	74	0	0
Neighbor	147	0	0
Other Relative	153	6	0.04

Table 6. Ratio of dream appearances of core support network members by relationship Type. This table compares the number of nights for all participants over a two-week period with members of participants' core support networks appearing in Dreams summarized by relationship type regardless of daily interaction. dream appearances were identified by researchers by comparing dream reports to individuals named by participants' in their support network. The column "not in dream" indicates the number of nights that the participants had a dream but no member of their support network of this relationship type appeared in the dream.

support ($\beta = 0.121$, $p = .004$). Eigenvector centrality had a large effect but more limited significance ($\beta = 1.427$, $p = .036$). Relationship length ($\beta = -0.196$, $p = .014$) was negatively associated with daily appearances.

For the participant-level trait variables, neuroticism was positive and significantly related to daily appearances, and openness was negative and significant. Attachment anxiety was negatively predictive of daily appearances. Notably, the differences between these effects and those found in the models in Sect. 2.2 indicate that controlling for within-relationship variance has a major effect on the model.

Relationship frequency remained a significant predictor of dream appearances ($\beta = 0.700$, $p < .001$), although its effect size was reduced compared to daily interactions. Closeness ($\beta = 0.268$, $p = .001$) and conflict ($\beta = 0.225$, $p = .002$) were again significant positive predictors. Eigenvector centrality ($\beta = 2.400$, $p = .013$) exhibited a stronger positive association with dream appearances than with daily interactions but was less significant. Unlike in daily appearances, financial support and relationship length were not significant.

For participant-level variables, extraversion and agreeableness were negative and significant for predicting dream appearances, while openness was positive and marginally significant ($p = .07$).

In comparing the two models, relationship frequency, closeness, and conflict consistently emerged as significant predictors of both daily interactions and dream appearances. Relationship frequency and closeness were more impactful for daily appearances.

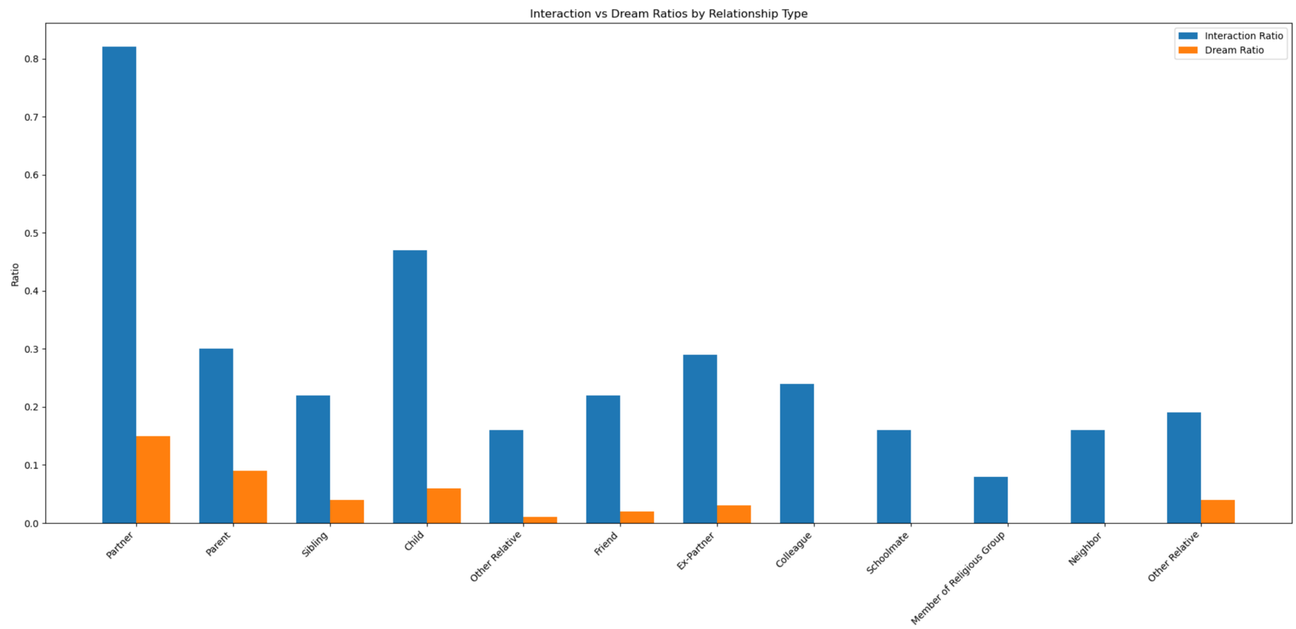

Fig. 2. Comparison of daily interaction and dream appearance ratios by relationship type. The chart displays the ratio of daily interactions (blue) and dream appearances (orange) across relationship types within participants' core support networks based on the results summarized in Tables 5 and 6.

Daily interactions and relationship type as predictors of dream appearances

Next we investigated how relationship type influenced the likelihood of dream appearances while controlling for daily interactions, as well as constructing interaction variables for the role of relationships in mediating dream appearance based on daily interaction (Table 8).

In the main-effects model, daily interactions at both the within-person ($\beta = 0.115$, $p = .03$) and between-person ($\beta = 0.622$, $p < .001$) levels are positive predictors of dream appearances. Likewise, family and close partners show consistently higher likelihoods of dream appearance, with particularly large effects for parents ($\beta = 2.393$) and partners ($\beta = 2.148$), followed by siblings ($\beta = 1.788$) and children ($\beta = 1.652$).

Interaction variables caused notable shifts in the model. While the main effects of daily interactions and relationship types remained significant, interactions between daily interactions and relationship types revealed complex dynamics. For within-persons daily interactions, the likelihood of sibling-related dream appearances decreased on days with higher interaction, while interactions with other relationship types did not show significant effects. For between-persons interactions, higher overall interaction levels with parents and siblings were associated with a reduced likelihood of their appearance in dreams.

For the participant-level variables, extraversion and agreeableness were negative and significant, while openness was positive and significant, similar to the previous set of models. Attachment avoidance was negative and significant in this model, indicating that there may be some kind of suppressor effect when daily interaction is incorporated.

Discussion

In an intensive longitudinal study of $N = 124$ participants from whom, across a 2-week period, we collected a total of 1,162 nights with recalled dream content (average 9.37 per participant), we found that the majority of dream nights for our participants involved strangers, and that many of them also did not involve identifiable individuals from their core support network or observed from diaries of daily life. This was true both in participant assessments of their dreams and in our own work matching daily logs and the core support network. This finding tracks with established findings that dreams include large numbers of strangers[32], and provides significant support for one of the core hypotheses of Social Simulation Theory, that strangers should be vastly overrepresented in dream content[19]. In the view of the SST laid out by Revonsuo et al.[19], strangers are overrepresented in dreams due to the evolutionary advantages of rapidly identifying familiar and unfamiliar individuals in the ancestral environment. In this context, humans largely lived in small, tight-knit social groups and strangers represented a direct threat to the group and thus were critical to quickly detect for survival. Of course, the participants in this study living in a contemporary setting encounter a wide variety of strangers in their daily lives, which may be mirrored in their dream experiences. This still does not explain why so many dreams would *only* include strangers, however, which aligns with the prediction of SST that strangers are overrepresented relative to their appearance in daily life.

Also in support of SST, personality was an important factor in the levels of dream content, with extraversion playing an important role in both increasing the number of friends dreamed about and decreasing the number of strangers, while openness led to higher levels of strangers, and neuroticism led to higher levels of relatives

						95% CI		95% Prob CI	
Outcome	Predictor	*β*	Prob	SE	Z-Stat	Low	High	Low	High
Daily Appearance									
	Level-1								
	Intercept	−1.313	0.212	0.073	−18.006	−1.456	−1.17	0.189	0.237
	Level-2								
	Relationship Length	−0.276**	0.431	0.083	−3.317	−0.439	−0.113	0.392	0.472
	Relationship Frequency	1.267***	0.78	0.093	13.555	1.084	1.45	0.747	0.81
	Closeness	0.443***	0.609	0.088	5.01	0.27	0.616	0.567	0.649
	Financial Support	0.335***	0.583	0.086	3.913	0.167	0.504	0.542	0.623
	Conflict	0.312***	0.577	0.075	4.166	0.165	0.459	0.541	0.613
	Eigen. Centrality	0.159*	0.54	0.075	2.134	0.013	0.306	0.503	0.576
	Level-3								
	BFI-E	0.01	0.503	0.093	0.108	−0.172	0.192	0.457	0.548
	BFI-A	−0.072	0.482	0.097	−0.737	−0.263	0.119	0.435	0.53
	BFI-C	0.035	0.509	0.087	0.4	−0.135	0.205	0.466	0.551
	BFI-N	0.213*	0.553	0.106	2.003	0.005	0.421	0.501	0.604
	BFI-O	−0.282**	0.43	0.085	−3.305	−0.449	−0.115	0.39	0.471
	AAS-ANX	−0.334***	0.417	0.09	−3.706	−0.511	−0.157	0.375	0.461
	AAS-AVOID	0.037	0.509	0.08	0.462	−0.119	0.193	0.47	0.548
Dream Appearance									
	Level-1								
	Intercept	−4.986	0.007	0.191	−26.082	−5.361	−4.611	0.005	0.01
	Level-2								
	Relationship Length	0.074	0.518	0.129	0.572	−0.179	0.327	0.455	0.581
	Relationship Frequency	0.65***	0.657	0.145	4.473	0.365	0.935	0.59	0.718
	Closeness	0.445**	0.609	0.144	3.088	0.162	0.727	0.541	0.674
	Financial Support	0.155	0.539	0.126	1.229	−0.092	0.403	0.477	0.599
	Conflict	0.23*	0.557	0.104	2.204	0.025	0.435	0.506	0.607
	Eigen. Centrality	0.215	0.554	0.113	1.906	−0.006	0.437	0.498	0.607
	Level-3								
	BFI-E	−0.417**	0.397	0.135	−3.077	−0.682	−0.151	0.336	0.462
	BFI-A	−0.278*	0.431	0.138	−2.013	−0.548	−0.007	0.366	0.498
	BFI-C	0.164	0.541	0.125	1.32	−0.08	0.409	0.48	0.601
	BFI-N	0.019	0.505	0.153	0.125	−0.281	0.319	0.43	0.579
	BFI-O	0.23	0.557	0.127	1.81	−0.019	0.48	0.495	0.618
	AAS-ANX	−0.016	0.496	0.134	−0.118	−0.279	0.247	0.431	0.561
	AAS-AVOID	−0.127	0.468	0.117	−1.084	−0.356	0.102	0.412	0.526

Table 7. Logistic mixed effects models predicting daily interaction and dream appearances from Relationship-Level Variables. These tables summarize the outputs of two fitted logistic mixed effects models predicting the presence of participant core support network members during the daytime and Dreams. models are fit at three levels to account for clustering with relationships and participants; observations within relationships (Level 1), relationship-level traits (Level 2), and participant traits (Level 3). BFI = Big five inventory: e = extraversion, a = agreeableness, c = conscientiousness, n = neuroticism, o = openness; aas = adult attachment scale: anx = anxiety, avo = avoidance. *$p < .05$, ** $p < .01$, *** $p < .001$. Prob. = The predicted probability from the log-odds coefficient.

appearing. Note that this set of findings strongly suggests that the simulations dreams produce are in service to pre-existing personality – related strivings and structure and are therefore functional. A previous study comparing the BFI dimensions to dream content found that extraversion, neuroticism, and openness were positively associated with incorporation from daily life[33]. This suggests an interesting new perspective on Social Simulation Theory; for those who are more outwardly-socially oriented, dreams serve as arenas for them to build up and reinforce relationships with acquaintances, while more inward-looking and open individuals experience their dreams to experiment with novel encounters with unknown individuals. These distinctions are also consistent with the cognitive version of the Continuity Hypothesis, as socially driven individuals are more likely to ruminate on new acquaintances and friends, and it will require more than this study to fully arbitrate between the two perspectives.

						95% CI		95% Prob CI	
Outcome	Predictor	*β*	Prob	SE	Z-Stat	Low	High	Low	High
Dream Appearance									
	Level-1								
	Intercept	−6.267	0.002	0.248	−25.24	−6.754	−5.78	0.001	0.003
	Daily Interaction(w)	0.115*	0.529	0.054	2.138	0.01	0.221	0.502	0.555
	Level-2								
	Daily Interaction(b)	0.622***	0.651	0.115	5.409	0.397	0.848	0.598	0.7
	Partner	2.148***	0.896	0.368	5.833	1.426	2.87	0.806	0.946
	Parent	2.393***	0.916	0.286	8.381	1.834	2.953	0.862	0.95
	Sibling	1.788***	0.857	0.356	5.024	1.09	2.485	0.748	0.923
	Child	1.652***	0.839	0.347	4.755	0.971	2.333	0.725	0.912
	Level-3								
	BFI-E	−0.366**	0.41	0.128	−2.848	−0.617	−0.114	0.35	0.472
	BFI-A	−0.305*	0.424	0.126	−2.412	−0.553	−0.057	0.365	0.486
	BFI-C	0.133	0.533	0.115	1.15	−0.093	0.358	0.477	0.589
	BFI-N	−0.049	0.488	0.145	−0.334	−0.333	0.236	0.417	0.559
	BFI-O	0.353**	0.587	0.123	2.877	0.113	0.594	0.528	0.644
	AAS-ANX	0.135	0.534	0.127	1.06	−0.115	0.385	0.471	0.595
	AAS-AVOID	−0.247*	0.439	0.112	−2.198	−0.467	−0.027	0.385	0.493
Dream Appearance									
	Level-1								
	Intercept	−6.243	0.002	0.249	−25.112	−6.731	−5.756	0.001	0.003
	Daily Interaction(w)	0.275*	0.568	0.112	2.451	0.055	0.494	0.514	0.621
	Level-2								
	Daily Interaction(b)	1.052***	0.741	0.195	5.384	0.669	1.435	0.661	0.808
	Partner	2.094**	0.89	0.647	3.238	0.827	3.362	0.696	0.966
	Parent	2.579***	0.929	0.286	9.008	2.018	3.14	0.883	0.959
	Sibling	1.823***	0.861	0.352	5.171	1.132	2.513	0.756	0.925
	Child	1.727***	0.849	0.405	4.261	0.933	2.521	0.718	0.926
	DI(w) * Partner	−0.035	0.491	0.188	−0.186	−0.402	0.333	0.401	0.582
	DI(w) * Parent	−0.264	0.434	0.143	−1.839	−0.545	0.017	0.367	0.504
	DI(w) * Sibling	−0.442*	0.391	0.223	−1.979	−0.88	−0.004	0.293	0.499
	DI(w) * Child	−0.115	0.471	0.179	−0.641	−0.467	0.237	0.385	0.559
	DI(b) * Partner	−0.404	0.4	0.35	−1.155	−1.09	0.282	0.252	0.57
	DI(b) * Parent	−1.003**	0.268	0.313	−3.207	−1.616	−0.39	0.166	0.404
	DI(b) * Sibling	−0.653	0.342	0.376	−1.736	−1.391	0.084	0.199	0.521
	DI(b) * Child	−0.478	0.383	0.305	−1.568	−1.075	0.12	0.254	0.53
	Level-3								
	BFI-E	−0.383**	0.405	0.126	−3.041	−0.63	−0.136	0.348	0.466
	BFI-A	−0.245	0.439	0.125	−1.955	−0.49	0.001	0.38	0.5
Continued									

						95% CI		95% Prob CI	
Outcome	Predictor	*β*	Prob	SE	Z-Stat	Low	High	Low	High
	BFI-C	0.178	0.544	0.114	1.566	−0.045	0.401	0.489	0.599
	BFI-N	0.007	0.502	0.143	0.049	−0.274	0.288	0.432	0.571
	BFI-O	0.319**	0.579	0.119	2.67	0.085	0.553	0.521	0.635
	AAS-ANX	0.123	0.531	0.124	0.992	−0.12	0.367	0.47	0.591
	AAS-AVOID	−0.269*	0.433	0.11	−2.441	−0.486	−0.053	0.381	0.487

Table 8. Logistic mixed effects main effects and interaction models predicting dream appearances from relationship type and daily appearances. These tables summarize the outputs of two fitted logistic mixed effects models predicting the presence of participant core support network members in Dreams as both a main effect of interaction during the day and an interaction model analyzing how daily interactions mediate the effects of relationship type at both within- and between-relationship levels. models are fit at three levels to account for clustering with relationships and participants; observations within relationships (Level 1), relationship-level traits (Level 2), and participant traits (Level 3). DI = daily interaction: $_W$ = within-relationship, $_b$ = between-relationship. BFI = Big five inventory: e = extraversion, a = agreeableness, c = conscientiousness, n = neuroticism, o = openness; aas = adult attachment scale: anx = anxiety, AVO = Avoidance. *p < .05, Prob. = the predicted probability from the log-odds coefficient.

An additional striking finding of our studies is that dreams involving the core support network and people from daily life are actually unusual simulations and constitute the clear minority of cases. The conspicuous lack of core support network characters in dreams suggests that dreams do not primarily serve as vehicles for reflecting on people who are most important to the dreamer or most involved in their life. This finding is particularly notable since diaries were completed just before bedtime, which should give us insight into the individuals with whom dreamers are most preoccupied from their previous day.

Schweickert, Xi, Viau-Quesnel, and Zheng[18] suggest that appearances of dream characters follow a power law distribution, in which characters who are more central and important in our lives are likely to more frequently appear in our dreams at a non-linear rate if they appear more than once. This idea builds on their previous work on the social networks of dreaming, which found that higher network centrality in a social network predicted dream appearance and that dream networks were less modular and more randomly organized than real-life social networks[34–36]. Our analysis offers novel context to this body of work. While eigenvector centrality was only marginally significant for dream appearance, this may be due to the relatively small size of the networks measured. Our observation that network centrality contributes to daily appearances and that these facilitate dream appearances offers a tentative mechanism for this process, as it may be that those who are more central in our cognitive social maps are more likely to be active parts of our daily life and this may lead to more frequent dreaming. Since factors like relational closeness and conflict predict dream appearances but other predictors like financial support did not, we contend that this supports the concept that our cognitive and emotional conceptions of others are key factors in stimulating dreaming about them, in addition to frequency of interaction. This observation is consistent with previous work finding that individuals often dream about individuals who are emotionally important but not part of our daily lives, such as ex-partners[37] and individuals who have died[38].

The results in Sect. 2.3.2 offer an intriguing follow-up to the isolation study by Tuominen, Olkoniemi, Revonsuo, and Valli[27] discussed in our introduction. In that study, individuals were more likely to dream about familiar individuals in isolation than out of isolation. Our interaction models found that for parents and siblings, daily engagement has a suppressing effect as a mediator in dream appearances on relationship types. This indicates that seeing a family member more regularly in daily life may actually diminish the likelihood of dreaming about them, and that this effect applies more to figures like parents and siblings than partners, although this observation may be influenced by the extremely high frequency of partner interaction in the study in relation to interaction with other relationship types. Consequently, dreams may serve a compensatory function for key attachment relationships, maintaining and solidifying bonds when the other person is absent. It is interesting to note that in the relationship-specific models openness predicted both lower levels of daily interaction with core support network members as well as higher levels of dream appearances. Openness may thus indirectly cause more active dreaming about relationships due to the tendency of open individuals to occupy more disconnected, open networks[39,40], although this remains a subject for future study. We suggest that the finding that extraversion and agreeableness are negatively associated with the appearance of core support network members in dreams likely corresponds with the result in our first study that indicates that extraversion leads to much higher levels of dreaming about friends. Extraversion may orient individuals more towards the cultivation of novel ties as well as the maintenance of more established ones. Since "weak" ties outside of one's core social network are often critical to social success, it may be that this type of dream content facilitates building these relationships for extraverted individuals[41]. We also note that extraverts are more likely to rapidly facilitate bonds with other extraverts[42], and future work could investigate whether or not extraverts are more likely to dream of each other as well in an early friendship stage.

In conclusion, we found that the majority of dreams involve strangers, and that personality and attachment style influence the population type of dream characters. We also found that relationship variables impact the appearance of close others in dreaming as well as daily interactions in waking life. For important family bonds, like parents and siblings, we found that daily interactions diminish dream appearances, indicating that dreams may serve a compensatory role for maintaining close emotional attachments and key relationships.

Our findings on the core support network are limited by the size of the network, as we only asked individuals to name a minimum of five and no more than eight important people. In addition, identifying characters from daily life in dreams required clear use of names or features and ambiguous cases were discarded, which means there may be more frequent dreaming of people from daily life than we could ascertain. In addition, the continuity, attachment, and social simulation hypotheses very often yield very similar predictions with regard to dream characters, therefore differentiating the predictions is certainly open to interpretations and will require more than a single study to tease out the differences. Another limitation is that we relied on self-report for ascertainment of medical diagnoses and lists of medications. We excluded individuals with existing neurologic diagnoses. Finally, our sample was a convenience sample of people interested in participating in studies on dreams and thus may not be representative of the general population. Future research should further investigate the role of strangers in dreams. As noted above, most characters in dreams are unknown to the dreamer. While this finding is consistent with some readings of the SST it is not obviously consistent with the attachment or the continuity hypotheses. In particular, future work could test the distinctions between dreams involving threatening and non-threatening strangers to see if their frequencies and intensities correspond to personality and attachment traits. In addition, while our work used an ego-network method for measuring social relationships, this could be expanded by an observational study on a group or groups that can measure network interactions from both sides of the relationships. We suggest that a good starting point would be measuring dyadic interactions between partners in the home compared to dream characterizations of each other over time.

Methods

We recruited volunteers who answered online adverts for a remote study on sleep, dreams, and nightmares. To be eligible, participants were required to: be at least 18 years old, reside in the US, speak and read English, have reliable Wi-Fi, and not have a current psychiatric or neurological diagnosis. An additional criteria of not having a current diagnosis of a sensitive skin condition was added during data collection after a few participants reported a negative but non-severe irritation from wearing the Dreem 3 headband[43], which was used to monitor sleep architecture. Eligibility criteria regarding psychiatric and neurological diagnosis was self-reported by participants and was not screened using questionnaires at the time of invitation, however those who scored severe or higher on any of the three categories of the Depression, Anxiety, Stress (DASS)[44] scale in a baseline survey were not invited to participate further in the study due to concerns about participant burden.

After completing a baseline survey including demographic questions on gender, ethnicity, and socio-economic status, volunteers ($N = 124$) were invited to participate in a two-week study in the home during which time they completed surveys every night and every morning. A randomized half of the participants also wore the Dreem 3 headband, as part of our overall research project[45,46]. We aimed to have participants contribute a maximum of 14 days and nights of surveys on sleep and dream measures described below. Surveys were distributed online using Qualtrics with unique identifying numeric codes. Participants received the first longitudinal survey link manually, then the system automatically sent subsequent links upon survey completion. Participants were instructed to complete the night surveys right before going to bed and the morning surveys right when they woke up on either their phone or computer. Researchers monitored survey submissions and reached out with survey links and reminders if participants did not submit at their usual times (participants were instructed to follow their normal sleeping schedule, so these times varied).

Ethics declarations

1. Institutional Committee that Approved the Experiment: National University Institutional Review Board, Study Number 2022-184-OTH, dated May 17th, 2022.
2. Confirmation of accordance with guidelines and regulations: This study was approved, and all methods were carried out in accordance with relevant guidelines and regulations for National University.
3. Informed consent Confirmation: Informed consent was received from all participants prior to participation.

IRB/Ethics: The study was overseen by the National University Institutional Review Board, study number 2022-184-OTH, dated May 17th, 2022, and informed consent was received prior to participation. Due to the remote nature of the study, participants were provided with a PDF of the consent form via email and met on Zoom with a researcher who read the consent letter aloud and answered any participant questions.

Sample characteristics ($N = 124$): Participant completion rates of surveys was 98%. Participants were on average 44.37 years old (SD = 14.93), predominantly female (69.4%) and White (66.9%). Over half of the participants had completed a Bachelor's degree or higher (67.7%), and over half had a household income of over $50,000 a year (65.3%).

Sample characteristics for support network analyses ($N = 121$): Participant completion rates of surveys was 98%. Participants were on average 44.31 years old (SD = 15.13), predominantly female (69.4%) and White (66.9%). Over half of the participants had completed a Bachelor's degree or higher (66.9%), and over half had a household income of over $50,000 a year (65.3%).

Baseline survey

Demographics: Participants answered questions on age, gender, ethnicity, education, income, and occupation.

Personality: To assess how personality affects different dimensions of spiritual beliefs or dream content and behavior, we utilized the well-validated Big Five Inventory (BFI)[47]. Specifically, we used the 60-item BFI-2, which has been shown to have strong factor structure in the main 5 dimensions as well as robust sub-facets for each factor.

Attachment: To measure trait attachment, we used the Revised Adult Attachment Scale for Close Relationships[48], an 18-item questionnaire that asks participants to rate their own similarity to a series of statements about relationships. These items were then combined into a two factor solution comprising the dimensions of anxiety and avoidance (see Supplementary Materials).

Ego-network: We gathered attachment network data following a similar approach to that in Berán, Pléh, Soltész, Rácz, Kardos, Czobor, and Unoka[49], which asked participants to name 5–8 members of their core support network and to rate each of these alters along several dimensions, such as conflict, trust, and shared values. We then also asked them to rate how close each alter was to the others, generating an alter-alter network that was used to calculate eigenvector centrality, which estimates centrality both on the nodes' own place in the network in addition to how well-connected their neighbors are[50]. After review, three participants were excluded from the analyses involving this support network due to illogical responses in the name generator (listing plurals like "my friends," etc.).

Longitudinal surveys

Daily activities: Every night, participants were asked to name the three activities that took the most time that day and to rate how they felt during those activities. In addition, they were asked about their three longest social interactions that day, what those interactions involved, and to rate how they felt during those interactions, which was lightly adapted from Reis, Sheldon, Gable, Roscoe, and Ryan Reis[51].

Dream collection: In the morning, participants were asked to report any dream content they could recall from the entire night. A later prompt asked participants to focus on the most impressive dream from the night and write it out in detail. They are then asked one final time for any additional content they recalled after completing other sections of the survey. In order to analyze dream content, these reports were cleaned for typos, abbreviations were expanded, and reports were separated into individual dreams. For dreams that had content reported multiple times, the reports were combined to include all relevant information. Any text not directly related to the dream experience was removed (for example "I had a dream that…" or "I think I remember that…" or "but in real life it's actually…") in order to avoid these words being included in the text analysis as part of the dream experience.

Each morning the participants were also asked to rate the content of their dreams in terms of mood and general themes using the structured Dreamland Questionnaire (DL-Q)[52]. This questionnaire asks participants a variety of questions about their dreams, including what kind of content appeared in them, the emotional tone of dreams, etc.

Identifying dream characters from daily life

Appearance of characters from participant support networks were assessed from dream narratives by comparing dream content with the names provided by participants. A positive match was only determined when the person could be unambiguously identified either by name or by an exclusive set (i.e. "my brothers" including a brother if they were a member of the support network). All dreams were first coded by a research assistant and then reviewed by two members of the research team. For people from everyday life, daily diaries were first reviewed and all identifiable individuals were coded if they were not also a member of the core support network. This list was then compared to the dream narratives to check for appearances in dreams by these individuals. Vague or unclear references were not counted (i.e. interactions during the day with "a cashier" and a dream about "a cashier" would not be a positive match).

Statistical analyses

The intensive diary longitudinal design of the study allowed for use of multilevel regression analyses (i.e., mixed-effects regression, random-coefficients modeling, hierarchical linear modeling). Multilevel regression techniques were developed to analyze nested, or hierarchical, data structures. The daily diary data and dream content assessments served as repeated measures that are nested within individuals, while interactions with members of participant support networks were further nested within each participant-alter relationship. Strengths of the multilevel-regression analytic procedures include: (a) capability of handling missing data and unbalanced designs (i.e., the number of assessment points and the timing of assessments can vary across participants); (b) addressing analytic issues that arises from aggregating over a large number of assessment occasions or not accounting for the nested structure of the data; (c) very efficient and powerful estimation procedures that utilize all data points available; and (d) modeling flexibility allowing for the inclusion of continuous or categorical, time invariant or time varying, predictors and covariates. In our models, we included individual participant variability as a random effect, which adjusts the intercepts based on participant variability. In the models dealing with participant-other relationships, we also incorporated intercepts for those relationships at a further level. All continuous longitudinal measures were centered either around the participant or relationship mean, which was then added to the model as a predictor at the level above. This is following the suggestion outlined in Hamaker and Muthén[53], who stress that inclusion of both the within- and between-persons predictors is critical for getting more accurate estimates, even when one of the pair is not statistically significant. To ascertain significance of model effects, we used Wald Confidence Intervals and Restricted Maximum Likelihood. Due to the large number of predictors in the bigger models, random-intercepts was chosen over a random-slopes approach to allow better interpretability of estimates between models. All analyses were run in Python using the pymer4 library, an extension of the lme4 R library for Python[54].

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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